

FEATURE LEARNING · DEPTH & HIERARCHY · TRAINING DYNAMICS · SCALING LAWS · SPIN GLASSES

SUMMARY

My research asks which structures make high-dimensional problems statistically or algorithmically hard, and which permit efficient learning or computation. In particular, I develop predictive theories of feature learning and the role of depth in neural networks, together with theories of dynamics, optimization, and statistical-to-computational gaps in spin glasses and random graphs. This work draws on mathematical tools from probability, optimization, learning theory, and statistical physics.

RESEARCH HIGHLIGHTS

FEATURE LEARNING

Asymptotics of feature learning in two-layer networks

Derived exact asymptotics for feature learning by conditioning on gradient-induced spikes, characterizing learned-feature spectra and generalization and yielding a new route to universality. Showed how batch size, repeated steps, and data reuse determine learnable directions, overturning fresh-batch hardness predictions and motivating a classification via symmetries and label transformations. Also characterized fundamental limits and statistical-to-computational gaps in multi-index models.

ICML 2024 Spotlight · AISTATS 2025 Oral · JMLR 2024 · ICML 2024 · AISTATS 2025

DEPTH & HIERARCHY

Depth as sequential dimension reduction

Led work identifying a computational advantage of depth on hierarchical targets: gradient descent learns successive representations by reducing effective dimension. Follow-ups extend the mechanism to compositional targets, random hierarchy models, scaling laws, and low-degree spectral filtering.

NeurIPS 2025 Spotlight · ICML 2026 · COLT 2026 · HiLD @ ICML 2026

SPIN GLASSES

Statics, dynamics, and statistical-to-computational gaps in spin glasses

Developed the dynamical cavity method into a rigorous analytical technique yielding asymptotically exact descriptions of first-order algorithms, and characterized stable local optima and phase transitions in random graphs and spin glasses, probing statistical-to-computational gaps.

COLT 2026 · arXiv 2025 · SLMath highlight

SELECTED PUBLICATIONS

Full list of publications: [Google Scholar](#)

NEURIPS

2025 · Spotlight

The Computational Advantage of Depth: Learning High-Dimensional Hierarchical Functions with Gradient Descent [arXiv](#)

Yatin Dandi, Luca Pesce, Lenka Zdeborová, Florent Krzakala

ICML

2024

The Benefits of Reusing Batches for Gradient Descent in Two-Layer Networks [arXiv](#)

Yatin Dandi, Emanuele Troiani, Luca Arnaboldi, Luca Pesce, Lenka Zdeborová, Florent Krzakala

COLT

2026

Rigorous Asymptotics for First-Order Algorithms Through the Dynamical Cavity Method [arXiv](#)

Yatin Dandi, David Gamarnik, Francisco Pernice, Lenka Zdeborová

PREPRINT

2025

Sequential Dynamics in Ising Spin Glasses [arXiv](#) · [SLMath](#)

Yatin Dandi, David Gamarnik, Francisco Pernice, Lenka Zdeborová

AISTATS

2025 · Oral

A Random Matrix Theory Perspective on the Spectrum of Learned Features and Asymptotic Generalization [arXiv](#)

Yatin Dandi, Luca Pesce, Hugo Cui, Florent Krzakala, Yue M. Lu, Bruno Loureiro

PNAS

2024

Sampling with Flows, Diffusion, and Autoregressive Neural Networks from a Spin-Glass Perspective [paper](#) · [arXiv](#)

Davide Ghio, Yatin Dandi, Florent Krzakala, Lenka Zdeborová

HiLD @ ICML

2026

Deep Learning as Neural Low-Degree Filtering: A Spectral Theory of Hierarchical Feature Learning [arXiv](#) · [code](#)

Yatin Dandi, Matteo Vilucchio, Luca Arnaboldi, Hugo Tabanelli, Florent Krzakala

AISTATS 2025	Fundamental Computational Limits of Weak Learnability in High-Dimensional Multi-Index Models paper Emanuele Troiani, Yatin Dandi , Leonardo Defilippis, Lenka Zdeborová, Bruno Loureiro, Florent Krzakala
ICML 2024 · Spotlight	Asymptotics of Feature Learning in Two-Layer Networks after One Gradient Step arXiv Hugo Cui, Luca Pesce, Yatin Dandi , Florent Krzakala, Yue M. Lu, Lenka Zdeborová, Bruno Loureiro
JMLR 2024	How Two-Layer Neural Networks Learn, One (Giant) Step at a Time paper · arXiv Yatin Dandi , Florent Krzakala, Bruno Loureiro, Luca Pesce, Ludovic Stephan

RESEARCH EXPERIENCE

SINCE 2022	PhD Researcher <i>EPFL · Advisors: Lenka Zdeborová and Florent Krzakala</i> - Develop predictive theories of feature learning in shallow and deep networks using high-dimensional probability, random matrix theory, and statistical physics. - Contribute more broadly to the mathematics of spin glasses and diffusion models, statistical-to-computational gaps, high-dimensional statistics, and random graphs.	LAUSANNE, SWITZERLAND
2024 – 2025	Visiting Researcher / Program Associate <i>MIT and Simons Laufer Mathematical Sciences Institute · Host: David Gamarnik</i> - Undertook an EPFL Doc.Mobility-funded visit at MIT and participated in SLMath’s program on Probability and Statistics of Discrete Structures. - Derived asymptotically exact dynamical equations for high-dimensional algorithms in the SK model and random graphs.	CAMBRIDGE AND BERKELEY, USA
SUMMER 2022	Research Intern <i>Mila · Advisors: Simon Lacoste-Julien and Yoshua Bengio</i> - Developed a theoretical framework for analyzing convergence of min-max algorithms, including the saddle-point Frank-Wolfe method. - Studied sampling and trajectory-inference algorithms motivated by applications in single-cell biology.	MONTREAL, CANADA
2019 – 2022	Undergraduate Researcher <i>IIT Kanpur and EPFL · Collaborators: Martin Jaggi, Piyush Rai, Abhishek Kumar</i> - Developed optimization methods for heterogeneous distributed learning, yielding work on implicit gradient alignment (AAAI 2022) and decentralized mixing (NeurIPS 2022 workshop). - Together with Abhishek Kumar (Google Brain) and Piyush Rai, developed generalized adversarially learned inference (AAAI 2021); also co-developed a joint image-video generation model (WACV 2020).	INDIA AND SWITZERLAND

EDUCATION

2022 – 2027	EPFL <i>PhD in Computer Science · expected 2027</i> Advisors: Lenka Zdeborová and Florent Krzakala.
2017 – 2022	Indian Institute of Technology Kanpur <i>BTech–MTech in Computer Science and Engineering</i> CPI: 9.6/10; Master’s CPI: 10/10. Thesis: <i>Tackling Heterogeneity in Collaborative Machine Learning</i> , co-supervised by Martin Jaggi and Piyush Rai.

RECOGNITION, TALKS, AND TECHNICAL DEPTH

RECOGNITION	EPFL AI Center PhD Fellowship, 2026 · Jane Street Graduate Research Fellowship (Honorable Mention), 2026 · EPFL Doc.Mobility Grant, 2024 · EDIC Fellowship, EPFL, 2022
SELECTED TALKS	HiLD Workshop, Banff, 2026 · Youth in High-Dimensions, Trieste, 2026 · ProbAI Scaling Laws, Warwick, 2026 · Physics of ML, Stanford, 2025 · SLMath Seminar, Berkeley, 2025 · JMM, Seattle, 2025 · Oberwolfach, 2023
TECHNICAL TOOLKIT	Python, C++ , PyTorch, JAX · NumPy, SciPy, scikit-learn · Linux, Git, Docker, LaTeX · Numerical linear algebra; distributed and federated optimization

SERVICE AND MENTORING. Reviewer for journals and conferences in machine learning, statistics, and applied mathematics; supervised several master’s-level research projects at EPFL.